Integration Theory MATM39: Summary of Important Results

Simon Gustafsson

Chapter 1

Sigma Algebras

Definition 1. *Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is called a* σ-algebra *on X if:*

(a) $X \in \mathcal{A}$,

(b) *if* $A \in \mathcal{A}$, *then* $A^c \in \mathcal{A}$,

(c) *if* $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, *then* $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$,

(d) *if* $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, *then* $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$.

Proof. Assume (a), (b), and (c). Let $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$. Then $A_i^c \in \mathcal{A}$ for all i, and by (c), $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{A}$. By (b) again, $\left(\bigcup_{i=1}^{\infty} A_i^c\right)^c \in \mathcal{A}$. By De Morgan's law,

$$\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i^c\right)^c \in \mathcal{A}$$

□

Proposition 1. *Let X be a set. Then the intersection of an arbitrary nonempty collection of σ-algebras on X is a σ-algebra on X.*

Proof. Let $\mathcal{C}$ be a nonempty collection of σ-algebras on X, and define

$$\mathcal{A} := \bigcap_{\mathcal{B} \in \mathcal{C}} \mathcal{B}.$$

Since every $\mathcal{B} \in \mathcal{C}$ contains X, we have $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Since each $\mathcal{B}$ is a σ-algebra, $A^c \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$, hence $A^c \in \mathcal{A}$. If $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\{A_i\}_{i=1}^{\infty} \subset \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Hence

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$$

for every $\mathcal{B} \in \mathcal{C}$, and therefore

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ-algebra on X. □

Corollary 1. *Let X be a set, and let $\mathcal{F}$ be a family of subsets of X. Then there exists a smallest σ-algebra on X that contains $\mathcal{F}$.*

Proof. Let $\mathcal{C}$ be the collection of all σ-algebras on X that contain $\mathcal{F}$. This collection is nonempty, since $\mathcal{P}(X)$ is a σ-algebra on X and contains $\mathcal{F}$.

Define

$$\sigma(\mathcal{F}) := \bigcap_{\mathcal{A}\in\mathcal{C}} \mathcal{A}.$$

By the previous proposition, $\sigma(\mathcal{F})$ is a σ-algebra on X. It contains $\mathcal{F}$, since every $\mathcal{A} \in \mathcal{C}$ contains $\mathcal{F}$. Moreover, if $\mathcal{B}$ is any σ-algebra on X that contains $\mathcal{F}$, then $\mathcal{B} \in \mathcal{C}$, and therefore

$$\sigma(\mathcal{F}) \subset \mathcal{B}.$$

Thus $\sigma(\mathcal{F})$ is the smallest σ-algebra on X containing $\mathcal{F}$. $\square$

Definition 2. *The* Borel σ-algebra *on $\mathbb{R}^d$ is the σ-algebra generated by the open subsets of $\mathbb{R}^d$. It is denoted by $\mathcal{B}(\mathbb{R}^d)$. Its elements are called* Borel sets.

Proposition 2. *The Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is generated by each of the following collections:*

(a) the collection of all closed subsets of $\mathbb{R}$,

(b) the collection of all intervals of the form $(-\infty, b]$,

(c) the collection of all intervals of the form $(a, b]$.

Proof. Let $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ denote the σ-algebras generated by the collections in (a), (b), and (c), respectively.

Since $\mathcal{B}(\mathbb{R})$ contains all open sets and is closed under complements, it contains all closed sets. Hence

$$\mathcal{B}(\mathbb{R}) \supset \mathcal{B}_1.$$

Every set $(-\infty, b]$ is closed, so $\mathcal{B}_1 \supset \mathcal{B}_2$. Moreover,

$$(a, b] = (-\infty, b] \cap (-\infty, a]^c,$$

so $\mathcal{B}_2 \supset \mathcal{B}_3$.

Finally, let $G \subset \mathbb{R}$ be open. For each $x \in G$, since G is open, there exist rational numbers $a, b \in \mathbb{Q}$ such that

$$x \in (a, b] \subset G.$$

Hence

$$G = \bigcup_{\substack{a,b\in\mathbb{Q} \\ (a,b]\subset G}} (a, b].$$

This is a countable union, since $\mathbb{Q}^2$ is countable. Therefore $G \in \mathcal{B}_3$. Thus every open set belongs to $\mathcal{B}_3$, so

$$\mathcal{B}(\mathbb{R}) \subset \mathcal{B}_3.$$

Therefore all four σ-algebras are equal. $\square$

Measures

Definition 3. *Let X be a set and $\mathcal{A}$ a σ-algebra on X. A function $\mu : \mathcal{A} \to [0,\infty]$ is called a* measure *if*

(a) $\mu(\emptyset) = 0$,

(b) for every $\{A_i\}_{i=1}^{\infty}$ *of pairwise disjoint sets in* $\mathcal{A}$, $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

Proposition 3. *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

(a) If $\{A_k\}$ is an increasing sequence in $\mathcal{A}$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{k\to\infty} \mu(A_k).$$

(b) If $\{A_k\}$ is a decreasing sequence in $\mathcal{A}$ and $\mu(A_n) < \infty$ for some n, then

$$\mu\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k\to\infty} \mu(A_k).$$

Proof. (a) Define $B_1 = A_1$ and $B_k = A_k \setminus A_{k-1}$ for $k \geq 2$. Then $\{B_k\}$ are disjoint and $\bigcup_k A_k = \bigcup_k B_k$. Hence

$$\mu\left(\bigcup_k A_k\right) = \sum_k \mu(B_k) = \lim_{k\to\infty} \sum_{i=1}^{k} \mu(B_i) = \lim_{k\to\infty} \mu(A_k).$$

(b) Since $\mu(A_n) < \infty$ for some n, we may replace $\{A_k\}$ by the tail $\{A_{k+n-1}\}$ and assume $\mu(A_1) < \infty$. Let $C_k = A_1 \setminus A_k$. Then $\{C_k\}$ is increasing and

$$\bigcup_k C_k = A_1 \setminus \bigcap_k A_k.$$

By (a),

$$\mu(A_1 \setminus \bigcap_k A_k) = \lim_{k\to\infty} \mu(C_k) = \lim_{k\to\infty} \big(\mu(A_1) - \mu(A_k)\big),$$

which gives $\mu(\bigcap_k A_k) = \lim_k \mu(A_k)$. □

Outer Measures

Definition 4. *Let X be a set. An* outer measure *on X is a function $\mu^* : \mathcal{P}(X) \to [0,\infty]$ such that*

(a) $\mu^*(\emptyset) = 0$,

(b) if $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$,

(c) for any sequence $\{A_n\}$ of subsets of X, $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Definition 5. *The* Lebesgue outer measure *on $\mathbb{R}$ is the function $\lambda^* : \mathcal{P}(\mathbb{R}) \to [0, \infty]$ defined by*

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) : \ A \subset \bigcup_{i=1}^{\infty} (a_i, b_i) \right\}.$$

Definition 6. *The* Lebesgue outer measure *on $\mathbb{R}^d$ is defined by*

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \mathrm{vol}(R_i) : \ A \subset \bigcup_{i=1}^{\infty} R_i \right\},$$

where each R_i is a bounded open rectangle in $\mathbb{R}^d$ and

$$\mathrm{vol}(I_1 \times \cdots \times I_d) = \prod_{j=1}^{d} |I_j|.$$

Definition 7. *Let μ^* be an outer measure on X. A set $B \subset X$ is called* μ^*-measurable *if for all $A \subset X$,*

$$\mu^*(A) = \mu^*(A \cap B) + \mu^*(A \cap B^c).$$

Remark 1. *Outer measures are defined on all subsets of X, but are only subadditive in general. Carathéodory's criterion selects a class of sets on which μ^* becomes additive. This allows us to restrict μ^* to a σ-algebra and obtain a genuine measure (in particular, the Lebesgue measure).*

Theorem 1. *Let X be a set and μ^* an outer measure on X. Let*

$$\mathcal{M}_{\mu^*} := \{B \subset X : B \text{ is } \mu^*\text{-measurable}\}.$$

Then:

(a) $\mathcal{M}_{\mu^*}$ is a σ-algebra,

(b) the restriction of μ^* to $\mathcal{M}_{\mu^*}$ is a measure.

Proof sketch. One shows that $\mathcal{M}_{\mu^*}$ is closed under complements directly from the definition. Closure under finite unions follows by applying the defining identity to B_1 and B_2 and decomposing sets appropriately.

Closure under countable unions is obtained by first proving the result for finite disjoint unions and then passing to the limit using subadditivity of μ^*.

Finally, countable additivity of μ^* on $\mathcal{M}_{\mu^*}$ follows by applying the defining property to disjoint measurable sets and using the previous decomposition. □

Remark 2. *Applying this theorem to the Lebesgue outer measure λ^* on $\mathbb{R}^d$ produces a σ-algebra $\mathcal{M}_{\lambda^*}$, called the* Lebesgue σ-algebra. *The restriction of λ^* to this σ-algebra is the Lebesgue measure. Thus, outer measure together with Carathéodory's criterion provides a systematic way to construct Lebesgue measure.*

Chapter 2

Chapter 3

Chapter 5